

Chapter 2

Variables and Datatype

A Variable is the name given to a memory location.

Ex.: $a = 30$ # Variables = container to store a value
 $b = \text{"harvy"}$ # keywords = reserved words in Python
 $c = 71.22$ # Identifiers = class / function / variable

* Datatype

- Integer : $a = 1$ # a is an Integer
 ~~$b = 5.22$~~
- Floating Point Number : $b = 5.22$ # b is a floating Point number.
- String : $c = \text{"Mahip"}$ # c is a string
- Boolean : $d = \text{False}$ # d is a boolean variable
- none : $e = \text{None}$ # e is a None variable

* Rules of Variables

- A variable name contain alphabets, digits and underscores.
- A variable name can only start with an alphabet and underscores.
- A variable name can't start with digit.
- No while space is allowed to be used.

Ex: Harvy, m18, -Seven, mahip_28 etc.

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* Operators in Python

- Arithmetic operators: $+$, $-$, $*$, $/$ etc

Ex: `a = 7`

`b = 4`

`c = 7 + 4`

7 and 4 are operands
and + is operator.

`print(c)`

- Assignment operators: $=$, $+=$, $-=$, etc.

Ex: `a = 4 - 2` # Assign 4-2 in a.

`print(a)`

`b = 6`

`b += 3`

increment the value of b by 3.

`b -= 3`

decrement the value of b by 3.

`print(b)`

Assign it to b.

- Comparison operators: $==$, $>$, $<=$, $<$, $!=$ etc.

Ex: `d = 5 > 5`

`print(d)`

always return boolean datatype

- Logical Operators: `and`, `or`, `not` etc.

Ex: `e = 1 or 0` # In or case if one is 1 then output
`print(e)`

* Type() Function and Type Casting

Type() function is used to find the datatype of a given variable in python.

Ex: `a = 31`

```
t = type(a)    # class <int>
print(t)
```

`b = "Manip" # class <str>`

```
t = type(b)
```

```
print(t)
```

A number can be converted into a string and vice versa (if possible)

There are many functions to convert one datatype into another.

`str(31) ⇒ "31" # integer to string`

`int("32") ⇒ 32 # string to integer`

* Input() Function

This function allows the user to take input from the keyboard as a string.

Ex: `A = input("Enter name")` # if a is "Harry"
The user entered Harry.

Note: It is important to note that the output of input is always a string even if you entered number. That's why we use type conversion.

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Practice Set

- 1) Write a Python Program to add two numbers.
- 2) Write a Python Program to find remainder when a number is divided by x .
- 3) Check the type of variable assigned, using `input()` function.
- 4) Use Comparison operator to find out whether a given variable a is greater than b or not.
Take $a = 34$ and $b = 80$.
- 5) Write a Python Program to find an average of two numbers entered by user.
- 6) Write a Python Program to calculate the square of a number entered by the user.